

4.3 Representing Functions as Power Series Worksheet

1. Find a power series representation of the function $g(x) = \frac{3x}{1-x^2}$ and state its interval of convergence.

We note that $g(x) = \frac{3x}{1-x^2}$ which resembles the sum of a geometric series with $a = 3x$ & $r = x^2$ and

therefore can be modeled by the power series $\sum_{n=0}^{\infty} 3x(x^2)^n = \sum_{n=0}^{\infty} 3x \cdot x^{2n} = \sum_{n=0}^{\infty} 3x^{2n+1}$ as long as

$$|x^2| < 1 \Rightarrow -1 < x < 1.$$

Therefore we can say that $g(x) = \frac{3x}{1-x^2} = \sum_{n=0}^{\infty} 3x^{2n+1}$ for x in the interval $(-1, 1)$.

2. Consider the series $\sum_{n=0}^{\infty} \frac{1}{3^n} x^n$. Find the function represented by this series. Determine the interval of convergence.

We see that $\sum_{n=0}^{\infty} \frac{1}{3^n} x^n = \sum_{n=0}^{\infty} \left(\frac{x}{3}\right)^n$. This is a geometric series with $a = 1$ & $r = \frac{x}{3}$ and so we have

$$\sum_{n=0}^{\infty} \frac{1}{3^n} x^n = \frac{1}{1 - \frac{x}{3}} = \frac{3}{3-x} \text{ for } \left|\frac{x}{3}\right| < 1 \Rightarrow -3 < x < 3$$

3. Consider the series $\sum_{n=0}^{\infty} 2^n x^n$.

- a) Find the function represented by this series. Determine the interval of convergence.

We see that $\sum_{n=0}^{\infty} 2^n x^n = \sum_{n=0}^{\infty} (2x)^n$. This is a geometric series with $a = 1$ & $r = 2x$ and so we have

$$\sum_{n=0}^{\infty} 2^n x^n = \frac{1}{1-2x} \text{ for } |2x| < 1 \Rightarrow -\frac{1}{2} < x < \frac{1}{2}.$$

- b) Use your knowledge from part (a) to find a function which represents the series $\sum_{n=1}^{\infty} 2n(2x)^{n-1}$ and state the radius of convergence.

We see that this new series is the derivative of the previous series from part (a). Therefore we know that if $\sum_{n=0}^{\infty} 2^n x^n = \frac{1}{1-2x}$ then $\sum_{n=1}^{\infty} 2n(2x)^{n-1} = \frac{d}{dx} \left[\frac{1}{1-2x} \right] = \frac{2}{(1-2x)^2}$ with a radius of convergence of $R = \frac{1}{2}$.

- c) Use your knowledge from part (b) to determine the exact sum of the series $\sum_{n=0}^{\infty} 2n \left(\frac{1}{4} \right)^{n-1}$.

We can see that this is the series from part (b) being evaluated when $x = \frac{1}{8}$. Since $x = \frac{1}{8}$ will be inside the interval of convergence for this series we can say that:

$$\sum_{n=0}^{\infty} 2n \left(\frac{1}{4} \right)^{n-1} = \frac{2}{\left(1 - 2 \left(\frac{1}{8} \right) \right)^2} = \frac{2}{\left(1 - \frac{1}{4} \right)^2} = \frac{2}{\left(\frac{3}{4} \right)^2} = \frac{32}{9}$$

4. Consider the series

$$\frac{2}{3} - \frac{10(x-1)}{9} + \frac{50(x-1)^2}{27} - \frac{250(x-1)^3}{81} + \frac{1250(x-1)^4}{243} + \dots$$

- a) For what values of x does the following series converge?

We need to start by writing this series using summation notation. We note that $\sum_{n=0}^{\infty} \frac{(-1)^n 2 \cdot 5^n (x-1)^n}{3^{n+1}}$

We could continue by using the Ratio Test, however may notice that if we attempt to rewrite the series a bit we discover that the series is geometric with $a = \frac{2}{3}$ and $r = \frac{5-5x}{3}$:

$$\sum_{n=0}^{\infty} \frac{(-1)^n 2 \cdot 5^n (x-1)^n}{3^{n+1}} = \sum_{n=0}^{\infty} \frac{2}{3} \cdot \left(-\frac{5}{3} \right)^n (x-1)^n = \sum_{n=0}^{\infty} \frac{2}{3} \left(-\frac{5(x-1)}{3} \right)^n = \sum_{n=0}^{\infty} \frac{2}{3} \left(\frac{5-5x}{3} \right)^n.$$

Therefore the series converges whenever

$$\left| \frac{5-5x}{3} \right| < 1 \Leftrightarrow \frac{|5-5x|}{3} < 1 \Leftrightarrow |5-5x| < 3 \Leftrightarrow -3 < 5-5x < 3 \Leftrightarrow -8 < -5x < -2 \Leftrightarrow \frac{2}{5} < x < \frac{8}{5}$$

- b) What is the sum of the series for these values of x ?

$$\text{Therefore, for } \frac{2}{5} < x < \frac{8}{5}, \text{ the series converges to } \frac{\frac{2}{3}}{1 - \left(\frac{5-5x}{3} \right)} = \frac{\frac{2}{3}}{\left(\frac{-2+5x}{3} \right)} = \frac{2}{5x-2}.$$

5. Consider the function $f(x) = \frac{2x}{2+x^2}$.

a) Find a power series representation for this function and state its interval and radius of convergence.

We note that this function resembles the sum of a convergent geometric series. We see that:

$$f(x) = \frac{2x}{2+x^2} = \frac{x}{1 - \left(-\frac{x^2}{2}\right)}. \text{ This function can now be expressed a geometric series with } a = x \text{ and}$$

$r = -\frac{x^2}{2}$ which converges when $\left|-\frac{x^2}{2}\right| < 1 \Leftrightarrow \frac{x^2}{2} < 1 \Leftrightarrow x^2 < 2 \Leftrightarrow -\sqrt{2} < x < \sqrt{2}$. Therefore, we have,

$$f(x) = \frac{x}{1 - \left(-\frac{x^2}{2}\right)} = \sum_{n=0}^{\infty} x \left(-\frac{x^2}{2}\right)^n \text{ whenever } -\sqrt{2} < x < \sqrt{2} \text{ (radius of convergence is } \sqrt{2} \text{)}.$$

b) Use information from part (a) to determine a power series representation for $g(x) = \frac{4-2x^2}{(2+x^2)^2}$ and state its radius of convergence.

$$\text{If we analyze } g(x) \text{ closely we see that } f'(x) = \frac{(2+x^2)2 - 2x(2x)}{(2+x^2)^2} = \frac{2x^2 - 4x^2 + 4}{(2+x^2)^2} = \frac{4-2x^2}{(2+x^2)^2} = g(x).$$

Therefore, if

$$f(x) = \sum_{n=0}^{\infty} x \left(-\frac{x^2}{2}\right)^n = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n x^{2n+1} \text{ then } f'(x) = g(x) = \sum_{n=1}^{\infty} \left(-\frac{1}{2}\right)^n (2n+1) x^{2n}$$

6. Suppose that $\sum_{n=0}^{\infty} a_n x^n$ is a power series with interval of convergence $(-1,1)$. Find the interval of

convergence for the power series $\sum_{n=0}^{\infty} a_n \left(\frac{x}{2}\right)^n$.

We note that $\sum_{n=0}^{\infty} a_n \left(\frac{x}{2}\right)^n$ will converge as long as $\frac{x}{2}$ is in the interval $(-1,1)$. $\frac{x}{2}$ is only in the this interval if x is in the interval $(-2,2)$. Therefore this series converges for all x in the interval $(-2,2)$.

7. Find a power series representation for $g(x) = \frac{1}{(1-x)(x-2)}$. Find the interval of convergence for power series.

We will first use partial fractions to write $g(x)$ as a sum of two fractions.

We know that $g(x) = \frac{1}{(1-x)(x-2)} = \frac{A}{(1-x)} + \frac{B}{(x-2)}$. It must be the case that $1 = A(x-2) + B(1-x)$.

$$\text{If } x=1, \text{ then } 1 = -1A \Rightarrow A = -1$$

$$\text{If } x=2, \text{ then } 1 = B$$

$$\text{Therefore } g(x) = \frac{1}{(1-x)(x-2)} = \frac{-1}{1-x} - \frac{1}{2-x} = \frac{-1}{1-x} - \frac{1/2}{1-\frac{x}{2}}$$

Each fraction represents the sum of a geometric series. Specifically, $\frac{-1}{1-x} = \sum_{n=0}^{\infty} -x^n$ for $-1 < x < 1$ and

$$\frac{1/2}{1-\frac{x}{2}} = \sum_{n=0}^{\infty} \frac{1}{2} \left(\frac{x}{2}\right)^n \text{ for } \left|\frac{x}{2}\right| < 1 \Rightarrow -1 < \frac{x}{2} < 1 \Rightarrow -2 < x < 2.$$

Therefore if we use values of x in the interval $(-1, 1)$ both series will converge and we can say that

$$\begin{aligned} g(x) &= \frac{1}{(1-x)(x-2)} = \frac{-1}{1-x} - \frac{1/2}{1-\frac{x}{2}} = \sum_{n=0}^{\infty} -x^n - \sum_{n=0}^{\infty} \frac{1}{2} \left(\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \left[-x^n - \frac{1}{2} \left(\frac{x}{2}\right)^n \right] \\ &= \sum_{n=0}^{\infty} \left[-x^n - \frac{1}{2} \cdot \frac{1}{2^n} x^n \right] = \sum_{n=0}^{\infty} \left(-1 - \frac{1}{2^{n+1}} \right) x^n \end{aligned}$$

8. Consider the series

$$\frac{3}{4} - \frac{3(x-3)}{8} + \frac{3(x-3)^2}{16} - \frac{3(x-3)^3}{32} + \frac{3(x-3)^4}{64} - \frac{3(x-3)^5}{128} + \dots$$

- a) For what values of x does the following series converge?

We need to start by writing this series using summation notation. We note that $\sum_{n=0}^{\infty} \frac{3(x-3)^n}{(-2)^{n+2}}$

We could continue by using the Ratio Test, however may notice that if we attempt to rewrite the series a bit we discover that the series is geometric with $a = \frac{3}{4}$ and $r = \frac{x-3}{-2} = \frac{3-x}{2}$:

$$\sum_{n=0}^{\infty} \frac{3(x-3)^n}{(-2)^{n+2}} = \sum_{n=0}^{\infty} \frac{3(x-3)^n}{4(-2)^n} = \sum_{n=0}^{\infty} \frac{3}{4} \left(\frac{x-3}{-2}\right)^n = \sum_{n=0}^{\infty} \frac{3}{4} \left(\frac{3-x}{2}\right)^n.$$

Therefore the series converges whenever

$$\left| \frac{3-x}{2} \right| < 1 \Leftrightarrow \frac{|3-x|}{2} < 1 \Leftrightarrow |3-x| < 2 \Leftrightarrow -2 < 3-x < 2 \Leftrightarrow -5 < -x < -1 \Leftrightarrow 1 < x < 5.$$

b) What is the sum of the series for these values of x ?

$$\text{Therefore, for } 1 < x < 5, \text{ the series converges to } \frac{\left(\frac{3}{4}\right)}{1 - \left(\frac{3-x}{2}\right)} = \frac{\frac{3}{4}}{\frac{x-1}{2}} = \frac{6}{4(x-1)}.$$

9. Consider the series $\sum_{n=0}^{\infty} \frac{5}{2} x^{3n+1}$.

a) Find a function $f(x)$ which is represented by this series and state the interval of convergence.

We note that $\sum_{n=0}^{\infty} \frac{5}{2} x^{3n+1} = \sum_{n=0}^{\infty} \frac{5}{2} x \cdot x^{3n} = \sum_{n=0}^{\infty} \frac{5}{2} x \cdot (x^3)^n$. This is a geometric series with $a = \frac{5}{2}x$ & $r = x^3$.

Therefore $\sum_{n=0}^{\infty} \frac{5}{2} x \cdot (x^3)^n = \frac{\frac{5}{2}x}{1-x^3}$ if $|x^3| < 1 \Rightarrow -1 < x < 1$. So the series $\sum_{n=0}^{\infty} \frac{5}{2} x^{3n+1}$ models the function

$$f(x) = \frac{\frac{5}{2}x}{1-x^3} \text{ on the interval } (-1, 1).$$

b) Show how your finding in part (a) can be used to show that $\sum_{n=0}^{\infty} \frac{5}{4 \cdot 8^n} = \frac{10}{7}$.

We note again that $\sum_{n=0}^{\infty} \frac{5}{2} x^{3n+1} = f(x)$ for x -values in the interval $(-1, 1)$. Since $x = \frac{1}{2}$ is inside of the

interval of convergence we know that $\sum_{n=0}^{\infty} \frac{5}{2} \left(\frac{1}{2}\right)^{3n+1} = f\left(\frac{1}{2}\right)$.

Through simplification we can see that $\sum_{n=0}^{\infty} \frac{5}{2} \left(\frac{1}{2}\right)^{3n+1} = \sum_{n=0}^{\infty} \frac{5}{2} \left(\frac{1}{2}\right) \left(\frac{1}{2^3}\right)^n = \sum_{n=0}^{\infty} \frac{5}{4 \cdot 8^n}$ and

$$f\left(\frac{1}{2}\right) = \frac{\frac{5}{2} \left(\frac{1}{2}\right)}{1 - \left(\frac{1}{2}\right)^3} = \frac{\frac{5}{4}}{\frac{7}{8}} = \frac{5}{4} \cdot \frac{8}{7} = \frac{10}{7}. \text{ Therefore we have } \sum_{n=0}^{\infty} \frac{5}{4 \cdot 8^n} = \frac{10}{7}.$$